

Daniel Rodríguez Carrión

R&D Engineer in Electronics, Embedded Systems and Advanced Sensing

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Professional Profile

R&D engineer specialized in **embedded cybersecurity** and certification of connected devices. Experience in IoT platform assessment, test automation, and validation of critical security mechanisms in Linux environments and low-power systems. Strong background in firmware, sensing, and edge computing, allowing me to connect development, validation, and compliance in embedded systems.

Professional Experience

Electronics and R&D Engineer

Quantum Aerospace 06/2026 – Present

- First engineer to join the electronics department, contributing to the technical definition and preliminary design of a mobile airport platform aimed at pilot assistance, obstacle detection, and relative positioning.
- Search for, analyze, and select sensors suitable for the project, evaluating different perception and detection technologies within aeronautical regulatory frameworks such as **DO-178** and **DO-278**.
- Started firmware development on the **NXP S32K344** microcontroller, focused on the embedded control architecture and the integration of system peripherals and communications.
- Evaluated an **i.MX8**-based ECU as a high-level processing unit, carrying out tests on the **Apalis Evaluation Board** with **Torizon OS**.

Cybersecurity and R&D Engineer

DEKRA

01/2025 – 02/2026

- Participated in the evaluation team for the **SESIP Level 5** certification of **Google pKVM for Android**, helping accelerate testing and delivery cycles through test automation.
- End-to-end management of **4 evaluation projects** on embedded and IoT platforms, all successfully completed, covering devices such as **nRF52833**, **RTL907XD**, **ST87M01**, and **Arm MPS3** with **RealSecuriTEE**.
- Covered the available network and communication interfaces of each device according to project scope, together with validation of critical mechanisms such as **secure boot**, **secure firmware upgrade**, **secure debug**, and **secure storage**.
- Developed tooling in **Python** and **C/C++**, including test automation using the **PSA API**, reducing delivery times for several test campaigns by **1–2 weeks**.
- Produced technical documentation and evidence packages for certification processes aligned with schemes such as **SESIP** and **RED**.

Mechatronics and Aerial Robotics Engineer (project-based collaboration, combined with DEKRA)

AVERON

01/2025 – 05/2025

- Technical collaboration in an aerial robotics startup, carried out alongside my main role in cybersecurity and R&D.
- Conceptual design and 3D modeling of components for UAV platforms using **CATIA V5**.
- Supported mechatronic integration and functional prototyping tasks for autonomous drone solutions.

Technical Staff and Researcher

University of Málaga

02/2022 – 09/2024

HortSost Project: [project page](#)

- Technical development in the **HortSost** project, working on sensing, firmware, electronics, and experimental validation over low-power IoT platforms, which became the technical foundation of my later specialization in embedded assessment and security.
- Deployed sensing solutions for environmental and energy monitoring, including **current**, **humidity**, **temperature**, **anemometry**, and **rainfall** sensors, together with **electronic signal conditioning** and **PCB design** tailored to each scenario.
- Designed and implemented a self-configuring wireless sensor network infrastructure for **edge computing**, based on **ESP32-C3**, **FreeRTOS**, **ESP-NOW**, and **MQTT**, with a coordinator node, auto-pairing, and Internet gateway capabilities.
- Experimentally evaluated the platform, achieving a **12.5%** reduction in compiled binary size with an optimized **ESP-IDF** configuration, enabling more efficient software deployment on **ultra-low-power** devices with a smaller memory footprint and more room for added functionality without penalizing execution times.
- Developed software for the **Acconeer A121** radar, focused on biometric readings such as **heart rate** and **respiration** for room climate regulation systems.

- Contributed to research results that led to a **paper and conference presentation** at the **Spanish Congress on Computer Science (CEDI 2024)**.

Technologies

Firmware and embedded	C, C++, ESP32, ESP8266, FreeRTOS, OTA, RISC-V
Automation and scripting	Python, Bash, CMake, Makefiles, test automation, PSA API
IoT and connectivity	MQTT, Node-RED, NB-IoT, BLE, Bluetooth Mesh, Thread, Zigbee, NFC
Networks and buses	Ethernet, MACsec, I2C, JTAG, PCIe
IoT protocols and services	HTTP/HTTPS, CoAP, LwM2M, TCP/IP, TLS/DTLS
Systems and infrastructure	Linux/Ubuntu, virtual machines, Raspberry Pi, Docker
Embedded security	Secure boot, secure storage, secure firmware upgrade, secure debug, applied cryptography
Design and prototyping	CATIA V5, 3D modeling, mechatronic integration

Education

- **MSc in Mechatronics Engineering** — University of Málaga, 2023
- **BSc in Electronics, Robotics and Mechatronics Engineering** — University of Seville / University of Málaga, 2021
- **Project Management for Photovoltaic Solar Installations** — Euroinnova, 2023

Certifications and Languages

- **CATIA V5 - Part Design Associate** — Dassault Systèmes, 2024
- **RPAS Pilot and Radio Operator** — Aerocámaras, 2019
- **Languages:** Native Spanish, English B2 (Cambridge FCE)

Publication

Self-configuring wireless sensor network for edge computing, Spanish Congress on Computer Science (CEDI), A Coruña, 2024.